90-Minute Hands-On Workshop

Build Your AI-Powered Digital COO & Research Assistant

The Lean Startup AI Assistant — validate an idea in under an hour, and build your own Claude Code commands & skills along the way.

Claude Code CLI · OpenCode CLI · Markdown · Git | Dr. Ibrahim Abualhaol · 18 July 2026 · ▢ QR → repo

SPEAKER NOTES

Room check before you start: everyone cloned the repo and bash `setup/verify-setup.sh` is all green. 90 minutes, hands-on — they leave with 6 commands and 1 skill in Git.

THE PROBLEM

Your kid is struggling. The appointment is in three weeks.

What do you say tonight?

SPEAKER NOTES

The hook. Say it, then **stop talking for three seconds** — the silence does the work. Roughly a third of any room has lived some version of this. No stat here on purpose: the old deck claimed "68% of founder time goes to ops" and that number had no source anywhere in the repo. A question needs no citation and lands harder.

THE IDEA

Ahmad is building the answer. Alone.

😵 Founder, alone

- Re-typing the same prompts
- Weeks lost to research & docs
- Guessing instead of measuring
- Drowning in the ops grind

🤖 Founder + AI-COO

- Generates interview scripts & landing copy in minutes
- Analyzes surveys & clinician interviews instantly
- Recommends pivot-or-persevere with evidence
- Drafts SOPs, models, email sequences

SPEAKER NOTES

Contrast, don't lecture. Left is their Tuesday; right is the rest of the workshop. 30 seconds. Bridge from the hook: a real founder is trying to answer that question, and he has no team.

LIVE DEMO

Let me show you something.

Switch to the terminal. Watch the AI read a messy CSV and reason in real time.

SPEAKER NOTES

Stop talking. Switch to the terminal now. Do not narrate the setup.

LIVE DEMO — RAW PROMPT

Messy CSV → market report in 30 seconds

```
claude "Read datasets/parent_survey_responses.csv (50 parents).
Analyze the data and produce a report: respondent summary, top 3
concerns, what they say they want most, willingness to pay, and key
verbatim quotes. Show the count behind every percentage.
Format as a professional markdown report."
```

The audience sees the AI read the file, reason, and write — no spreadsheet, no analyst.

SPEAKER NOTES

THE most important moment of the workshop (FACILITATOR-CHECKLIST.md). Rehearse until flawless. Use the real CSV from the repo. They must *feel* messy-CSV-to-report in ~30s. This earns you the next 85 minutes.

LIVE DEMO — THE COMMAND 🔧

Same output. One-tenth the typing.

```
/project:analyze-survey datasets/parent_survey_responses.csv
```

Identical report — from a reusable command you'll learn to build today.

Today you won't just use prompts. You'll build tools.

SPEAKER NOTES

The turn: same output, one line. This is the whole thesis — they are not here to use prompts, they are here to build tools. Copy reference-commands/analyze-survey.md into .claude/commands/ beforehand.

THE FRAMEWORK

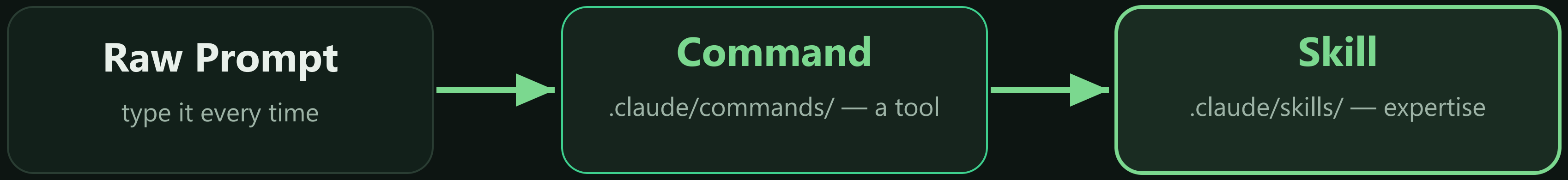
The Lean Startup Loop — with AI at every stage

SPEAKER NOTES

Build-Measure-Learn maps 1:1 onto Modules 1-3. Point at each box and name the module.

THE PROGRESSION

Prompt → Command → Skill



Modules 1–3 build commands. Module 4 bundles them into a skill.

SPEAKER NOTES

The spine of the day. Modules 1-3 build commands, Module 4 bundles them into a skill. Come back to this diagram at every module boundary.

THE RUNNING CASE STUDY

Meet Ahmad 🌱

Healthcare data engineer building Kinsight — connecting parents and clinicians between appointments.

Stage: Pre-revenue, pre-product
Budget: $8,000 bootstrap
Team: Solo. No clinicians, no analyst, no COO.
Why him: 10 years of EHR plumbing — and a daughter who waited 5 months.

His week threads all four modules.

SPEAKER NOTES

Ahmad threads all four modules. Full profile: case-study/founder-profile.md. Keep him concrete — solo, $8k, pre-revenue. The detail that matters: he's an **engineer**. That's why his bet on slide 12 is a score, and it's why the data is about to hurt.

ROADMAP

Today — 90 minutes

Time	Module	You build 🔧
0:00–0:12	Opening — the overwhelmed founder	—
0:12–0:35	Module 1 · BUILD	`/project:hypothesis` (guided)
0:35–0:55	Module 2 · MEASURE	2 analysis commands (semi-guided)
0:55–1:15	Module 3 · LEARN	2 decision commands (independent)
1:15–1:25	Module 4 · OPERATE	`lean-startup` skill (capstone)
1:25–1:30	Closing — next steps	—

SPEAKER NOTES

Timings match README.md. Say out loud: 'the wrench means you build something permanent.' Press T to start the workshop timer.

MODULE 1 · BUILD

Before you build, hypothesize

> **"We believe [segment] has [problem] and will [behavior] if we offer [solution]."**

A hypothesis is a bet you can test — specific, falsifiable, cheap to check.

SPEAKER NOTES

The format is falsifiable on purpose. A hypothesis you cannot lose is not a hypothesis.

MODULE 1 · BUILD

Ahmad's starting hypothesis

"We believe parents of 11-17 year olds will pay a monthly subscription for an AI risk score that tells them how their teen is really doing."

Riskiest assumption: that a *score* is what parents want. (Spoiler: the data disagrees.)

SPEAKER NOTES

His bet is WRONG and the data kills it in Module 3 — plant that now, do not reveal it. Source: case-study/hypothesis_v1.md. Ask the room: "who thinks this is a good idea?" Plenty of hands. It sounds obviously right, which is the whole point.

EXERCISE 1A · HANDS-ON

Generate 3 hypotheses

```
claude "I'm building a platform that helps parents and child
mental-health clinicians support a young person between appointments.
Generate 3 testable Lean Startup hypotheses in the format: 'We believe
[segment] has [problem] and will [behavior] if we [solution].'
Output a numbered markdown list with a rationale each."
```

Using your own idea? Swap the first sentence.

SPEAKER NOTES

Exercise 1A — exercises/module-1-build/exercise-1a-hypotheses.md. ~4 min. Walk the room. Anyone with their own idea swaps the first sentence.

🔧 **ANATOMY OF A COMMAND**

A command is just a markdown file

```
# .claude/commands/hypothesis.md
You are a Lean Startup strategist.
Generate 3 testable hypotheses for:

$ARGUMENTS

Format each as "We believe...".

Project context:
!cat CLAUDE.md
```

1. Filename → command name

`hypothesis.md` → `/project:hypothesis`

2. $ARGUMENTS

whatever you type after the command

3. !cat — shell injection

live file contents, no copy-paste

SPEAKER NOTES

Slow down. This is the first 🔧 beat. Three ideas only: filename becomes the command, $ARGUMENTS takes input, !cat injects live context. Do not add a fourth.

🔧 EXERCISE 1A-CMD · GUIDED

Build your first command

```
cat > .claude/commands/hypothesis.md << 'EOF'
You are a Lean Startup strategist helping an early-stage founder.
Generate 3 testable hypotheses for the following startup idea:

$ARGUMENTS

Format each as "We believe [segment] has [problem] and will
[behavior] if we offer [solution]."

For each, also include:
- Riskiest Assumption — the belief that, if wrong, sinks it
- Cheapest Test — the fastest way to validate it
```

Test: `/project:hypothesis a parent-clinician mental health platform`

SPEAKER NOTES

Exercise 1A-cmd — GUIDED, you walk every step. ~2 min. Windows users: the PowerShell here-string is in the exercise file. Answer key: reference-commands/hypothesis.md.

EXERCISE 1B · HANDS-ON

Customer interview script

```
claude "Based on this hypothesis: 'parents will pay a subscription
for an AI risk score about their teen,' generate a 10-question customer
discovery interview script. Include warm-up, pain-point, solution-fit,
and willingness-to-pay questions. Format as markdown."
```

The Mom Test rule: ask about their life and past behavior — never pitch.

SPEAKER NOTES

Exercise 1B — exercise-1b-interview-script.md. The Mom Test rule matters more here than anywhere: ask a worried parent "would you want to know if your child was at risk?" and 100% say yes. You've learned nothing. Ask what they actually did at 2am. Also: they interview **parents**, never children.

EXERCISE 1C · HANDS-ON

Landing page brief

```
claude "Write a landing page copy brief for a platform connecting
parents and child mental-health clinicians. Include: 1 headline (max 8
words), 1 subheadline (max 20 words), 3 value props with icons, and
1 CTA button. Target: parents of 11-17 year olds.
Tone: calm, credible, never alarmist."
```

This landing page is your Build-phase experiment — Module 2 measures it.

SPEAKER NOTES

Exercise 1C — exercise-1c-landing-page.md. **Watch what the AI reaches for.** The highest-converting headline here is fear — "Would you know if your child was in crisis?" — and it's a lie about a medical capability aimed at frightened people. If someone's draft does that, put it on screen: finding the line is the exercise. See the rejected copy in sample-outputs/landing_page_brief_example.md.

SHOW & TELL

What did you build?

2-3 volunteers share their output — and their first command file.

SPEAKER NOTES

2-3 volunteers. Ask to see the command FILE, not just the output — that is the graduation moment.

MODULE 1 · RECAP

In 23 minutes you produced...

- ✅ A tested hypothesis
- ✅ A 10-question interview script
- ✅ A landing page brief
- ✅ Your first reusable command 🔧

SPEAKER NOTES

23 minutes, four artifacts. Name the shift explicitly: prompt user -> toolmaker.

MODULE 2 · MEASURE

Data beats opinions.

Let AI be your research analyst.

SPEAKER NOTES

Quote slide. Let it sit. Module 2 starts here (0:35).

Signal vs. noise

Parent survey

50 parents — what they say they need.

Clinician interviews

20 clinicians — the other side of the gap.

Competitive scan

5 platforms — what's already crowded.

New pattern this module: commands that take a file as input via `!cat $ARGUMENTS`.

SPEAKER NOTES

New pattern this module: commands that take a FILE as input via !cat $ARGUMENTS. Flag it now, they build it on slide 24.

Ahmad's data landscape

Source	What it is	Signal type
`competitors.md`	5 platforms, prices, features	Competitive intel
`parent_survey_responses.csv`	50 parent responses	Demand + willingness-to-pay
`clinician_interviews.json`	20 clinician interviews	The other side of the gap

Two populations, two files. Nobody surveyed the kids — parents and clinicians report, minors don't.

SPEAKER NOTES

All three files are real and in the repo. Data dictionary: datasets/data_dictionary.md. Worth saying out loud: the datasets are synthetic, and there is deliberately **no data attributed to a young person**. That's a design choice they should copy in their own research.

EXERCISE 2A · HANDS-ON

Competitive analysis

```
claude "Read case-study/competitors.md. Create a competitive
analysis matrix (Platform, Price, Differentiator, Target User,
Parent/Clinician Reach, Gap). Then write a 200-word landscape summary
and identify 2 underserved niches."
```

3 of 5 already ship a score

0 of 5 connect parent → clinician

SPEAKER NOTES

Exercise 2A — exercise-2a-competitive-analysis.md. The setup for Thursday: **3 of 5 rivals already ship a score**, so even if parents wanted it, it wouldn't differentiate him. 0 of 5 own the handoff. Don't reveal the survey yet — let the two signals land separately.

🔧 EXERCISE 2A-CMD · SEMI-GUIDED

Fill in the skeleton

```
You are a competitive intelligence analyst.
Read the following competitor data:
!cat $ARGUMENTS

## Your Task
[PARTICIPANTS FILL THIS IN]

## Output Format
1. Comparison matrix: [YOU DEFINE COLUMNS]
2. 200-word landscape summary
3. 2 underserved niches
```

`!cat $ARGUMENTS` → works on any competitor file.

SPEAKER NOTES

Exercise 2A-cmd — SEMI-GUIDED. The ramp starts: you give the skeleton, they fill the task and columns. Do not solve it for them.

EXERCISE 2B · HANDS-ON

Analyze the survey (50 rows)

```
claude "Read datasets/parent_survey_responses.csv (50 parents).
Report: respondent summary, top 3 concerns, what they say they want
most, willingness to pay, key verbatim quotes. Show the count behind
every percentage. Professional markdown."
```

76% would refuse a risk score

64% just want to know what to say

56% pay $5-10, only 14% pay $20+

SPEAKER NOTES

Exercise 2B — exercise-2b-survey-analysis.md. These numbers are exact row counts of the 50-row CSV, not decoration: 38/50, 32/50, 28/50, 7/50. Let them recompute. **This is where Ahmad's bet dies** — but don't say it yet, that's slide 30.

🔧 EXERCISE 2B-CMD · SEMI-GUIDED

Build the survey command

```
cat > .claude/commands/analyze-survey.md << 'EOF'
You are a market research analyst.
Analyze the following survey data:
!cat $ARGUMENTS

Report: respondent summary · top 3 concerns · stated top need ·
willingness to pay · verbatim quotes · 3 "So What?" insights.
[YOU ADD: rules — show the counts, flag thin data,
and say so when preference and willingness-to-pay disagree]
EOF
```

Test: `/project:analyze-survey datasets/parent_survey_responses.csv`

SPEAKER NOTES

Exercise 2B-cmd — semi-guided. They add the rules themselves. Answer key: reference-commands/analyze-survey.md.

TECHNIQUE

The "So What?" follow-up + chaining

Iterate

```
claude "Based on that analysis, give me
exactly 3 actionable insights. For each:
evidence, implication, next step."
```

Chain 🔧

The output of `analyze-survey` becomes the input to next module's `pivot-decision`. Commands chain — mirroring the loop itself.

SPEAKER NOTES

Chaining preview: analyze-survey output becomes pivot-decision input. The commands chain the same way the loop does.

MODULE 2 · RECAP

In 20 minutes...

- ✅ Competitive matrix
- ✅ Survey report
- ✅ Actionable insights
- ✅ 2 reusable analysis commands 🔧

SPEAKER NOTES

20 minutes, two analysis commands. Halfway. Check the room's energy here.

MODULE 3 · LEARN

Pivot, persevere, or zoom in?

PERSEVERE

Data supports the bet. Keep going, optimize.

PIVOT

Core thesis is wrong. Change direction.

ZOOM IN

Thesis right, lever wrong. Narrow to what works.

SPEAKER NOTES

Module 3 (0:55). Three options, and ZOOM IN is the one founders forget exists.

Ahmad's dilemma

His bet

An AI risk score is the hero feature. Parents subscribe.

His data

76% would refuse a score about their own child. Only **12%** rank one as their top need. **64%** just want to know what to say.

The data — not his engineering instinct — makes the call.

SPEAKER NOTES

The tension lands here: his bet vs. his data. Do not resolve it yet — let them sit in it. The human beat: he's spent ten years being the person who measures things, and 38 out of 50 parents just told him not to measure their kid.

EXERCISE 3A · HANDS-ON

Synthesize & decide

```
claude "Acting as my strategic advisor. HYPOTHESIS: parents pay a
subscription for an AI risk score. PARENTS (n=50): 76% would refuse a
score; 12% rank one as their top need; 64% want to know what to say;
56% pay $5-10, 14% pay $20+. CLINICIANS (n=20): 85% say intake is the
top pain; 70% would pay per seat; 0/20 get between-session signal.
MARKET: 3/5 rivals ship a score; 0/5 connect parent to clinician.
Should I PIVOT, PERSEVERE, or ZOOM IN? Structure:
Recommendation · Evidence · Revised Hypothesis · Risk · Confidence."
```

SPEAKER NOTES

Exercise 3A — exercise-3a-pivot-or-persevere.md. Every number in this prompt is a real row count from the repo's data. Push the room: **what can this survey NOT tell him?** Answer: whether a score would actually work. 50 parents is a market signal, not clinical evidence.

🔧 **EXERCISE 3A-CMD · INDEPENDENT**

Build it yourself — hints only

- Filename: `.claude/commands/pivot-decision.md`
- Use $ARGUMENTS for the hypothesis under review
- Use `!cat` to inject your Module-2 data files
- Output: Recommendation · Evidence · Revised Hypothesis · Risk · Confidence
- Think: what context does the AI need to decide well?

No skeleton. You've built three commands — you've got this.

SPEAKER NOTES

Exercise 3A-cmd — INDEPENDENT. Hints only, no skeleton. They have built three commands; let them struggle productively. reference-commands/pivot-decision.md if truly stuck.

AHMAD'S AI RECOMMENDATION

ZOOM IN

Revised hypothesis: clinicians will pay a per-seat licence for an intake summary built from the parent's own observations — and parents will sustain it because they get back a conversation guide, not a score.

Confidence: HIGH — three independent signals agree (parents reject it · rivals already have it · clinicians will pay for the alternative). The payer moved, and the riskiest feature is gone.

The data didn't say "wrong feature." It said don't build the diagnostic.

SPEAKER NOTES

The payoff — **slow down here**. ZOOM IN, not pivot: thesis right, lever wrong. Land the second lesson explicitly: the riskiest thing to build was also the thing users didn't want and he couldn't responsibly validate. In health those two arrive together. Listening to customers and behaving well pointed the same way — that's luck worth naming, not a law. Full reasoning: case-study/pivot_decision.md.

EXERCISE 3B · HANDS-ON

Generate a 2-week sprint plan

```
claude "Based on the zoom-in to 'clinician intake summary from
parent observations, no risk score,' create a 2-week sprint plan:
updated hypothesis, 5 prioritized tasks (solo founder), success
metrics, bootstrap budget under $1,000, risks + mitigations.
Markdown with a Gantt-style text timeline."
```

The sprint tests the scary thing: will 10 families still check in on day 14?

SPEAKER NOTES

Exercise 3B — exercise-3b-sprint-plan.md. Constraints are the point: solo founder, under $1,000. If their plan opens with "build the MVP," push back — the whole sprint is a manual SMS pilot with 10 families and zero code. You can fake the product; you cannot fake whether people show up on day 14.

🔧 EXERCISE 3B-CMD · INDEPENDENT

Build it yourself — hints only

- Filename: `.claude/commands/sprint-plan.md`
- $ARGUMENTS = the strategic direction / pivot
- Bake in your constraints (solo, <$1,000) so you never re-type them
- Include: tasks · timeline · budget · metrics · risks

Test: `/project:sprint-plan "Pivot to transparency-focused brand"`

SPEAKER NOTES

Exercise 3B-cmd — independent again. Baking constraints into the command means never re-typing them.

THE COMPLETE LOOP

Your command chain = the B-M-L loop

`hypothesis` → `competitive-analysis` / `analyze-survey` → `pivot-decision` → `sprint-plan`

A complete methodology, codified — and it travels in Git.

SPEAKER NOTES

Their four commands ARE the loop. This is the slide that makes the toolkit feel like a system.

MODULE 3 · RECAP

In 20 minutes...

- ✅ Pivot decision (evidence-based)
- ✅ 2-week sprint plan
- ✅ Revised hypothesis
- ✅ 2 decision commands — built independently! 🔧

SPEAKER NOTES

20 minutes, and they built both commands solo. Say that out loud — they earned it.

MODULE 4 · OPERATE

Don't hire a COO. Prompt one.

SPEAKER NOTES

Module 4 (1:15). The line the workshop is named for. Pause.

MODULE 4 · OPERATE

From prompts → commands → skills

Commands = tools

You pick them up and invoke them. Single tasks.

Skills = expertise

Claude carries them and applies them automatically across a whole domain.

SPEAKER NOTES

Tools vs. expertise. One sentence each, then move.

Commands vs. Skills

Dimension	Commands	Skills
Trigger	Manual — `/project:x`	Auto — context-detected
Scope	One task	A whole domain
Form	One markdown file	Folder + SKILL.md + files
Best for	Repetitive on-demand tasks	Expertise you always want
Analogy	A tool in your toolbox	A skill in your brain

SPEAKER NOTES

The comparison table people photograph. Give them 15 seconds to shoot it.

EXERCISE 4A · HANDS-ON

SOP generator

```
claude "Create an SOP for onboarding a new clinic onto the
platform. Include: title + version, purpose/scope, step-by-step process
with decision points, email templates, escalation criteria, and KPIs.
Professional markdown."
```

Not this way: the safeguarding SOP. That one is written with a clinician and a lawyer — never a prompt.

SPEAKER NOTES

Exercise 4A — exercise-4a-sop-generator.md. Time is short; copying the reference command is fine. **Do not skip the amber box** — it's the most important 20 seconds of Module 4. AI is great for ops docs that are cheap to fix. It is the wrong tool where being confidently wrong hurts someone and you can't tell. An LLM will draft a beautiful safeguarding procedure that's wrong in ways a founder cannot see, and it'll sit in the repo looking official.

🔧 EXERCISE 4B · CAPSTONE

Build the lean-startup skill

```
mkdir -p .claude/skills/lean-startup/templates

cat > .../SKILL.md << 'EOF'
---
name: lean-startup
description: Lean Startup toolkit for
  Build-Measure-Learn cycles.
---
# Lean Startup Skill
## When to Activate ...
## Methodology ...
## Principles ...
EOF
```

```
.claude/skills/lean-startup/
├── SKILL.md
├── templates/
└── examples/
```

Now Claude auto-activates your methodology whenever you open this project.

SPEAKER NOTES

The CAPSTONE — exercise-4b-capstone-build-skill.md. Folder + SKILL.md + templates + examples. Full bundle: reference-skills/lean-startup/.

POWER TIPS

Four force multipliers

🔗 Chain

Run the whole B-M-L loop, command by command.

🧠 Context

`!cat CLAUDE.md` in every command = shared brain.

📦 Version

`git add .claude/` — your toolkit ships with the repo.

⚡ Batch

"For each company in competitors.md, generate a one-page brief and save it."

SPEAKER NOTES

Four multipliers, ~2 min total. Details in exercises/module-4-operate/power-tips.md.

MODULE 4 · RECAP

In 10 minutes...

- ✅ A complete SOP
- ✅ The lean-startup skill 🔧
- ✅ A complete, Git-shareable AI-COO toolkit

SPEAKER NOTES

10 minutes, and the toolkit is complete and Git-shareable.

CLOSING

Ahmad's week — before & after

Day	Produced	Built 🔧	Time
Mon	Hypotheses, script, landing	`hypothesis`	15 min
Wed	Competitive + survey, insights	2 analysis cmds	20 min
Thu	Pivot decision, sprint plan	2 decision cmds	10 min
Fri	SOPs, dashboard	SOP cmd + skill	10 min

~55 min AI-assisted vs. 2-4 weeks manual — and he killed his own best idea on Thursday.

SPEAKER NOTES

Closing (1:25). ~55 min AI-assisted vs 2-4 weeks manual — say the number. Then the real point: the speed isn't the win. **The win is that he found out on Thursday instead of after six months of building the score.**

CLOSING

Your `.claude/` folder

```
.claude/
├── commands/    ← 6 tools you invoke on demand
│   ├── hypothesis.md
│   ├── competitive-analysis.md
│   ├── analyze-survey.md
│   ├── pivot-decision.md
│   ├── sprint-plan.md
│   └── generate-sop.md
└── skills/
    └── lean-startup/    ← expertise Claude carries
```

This is your AI-COO's brain. It lives in Git.

SPEAKER NOTES

This is the artifact they keep. 'Your co-founder git clones and gets your whole AI-COO on day one.'

CLOSING

5 key takeaways

- **AI is a force multiplier** — it amplifies your speed, not your judgment
- **Framework + AI = superpower** — the loop gives AI direction
- **Start tonight** — pick ONE task, prompt your AI-COO
- **Build tools, not just prompts** — commands & skills compound
- **Your .claude/ folder is your startup's operational brain**

SPEAKER NOTES

Five takeaways. If you are over time, cut to the last two — they are the ones that matter.

HOMEWORK

Tonight: pick ONE task. Build a command for it.

SPEAKER NOTES

The single sentence you want them to leave with. Say it, stop, do not add to it.

RESOURCES

Take it with you

📦 The repo

`github.com/alhaol/ai-powered-digital-coo-research-assistant`

Commands, skill, datasets, exercises, playbook — the workshop lives in `workshop-repo/`.

📖 Keep learning

- Claude Code docs — commands & skills
- *The Lean Startup*, *The Mom Test*, *Zero to One*
- Take-home booklet PDF

Your commands & skill are local — `git push` to save them.

SPEAKER NOTES

Repo is public: `github.com/alhaol/ai-powered-digital-coo-research-assistant` — display the QR here. Remind them the workshop files are under `workshop-repo/`. Booklet + docs links go in the 24h follow-up email.

THANK YOU

Questions?

Don't hire a COO. Prompt one.

Dr. Ibrahim Abualhaol · AI Technical Lead · Ottawa, Canada · 18 July 2026

SPEAKER NOTES

Q&A. Last line: their commands and skill are local until they git push. Send them home with that.